1 Litre Water Bottle – 3D Design (Fusion 360)

Introduction

This project involves the **3D modeling of a 1-litre water bottle** using **Autodesk Fusion 360**. The objective of the project was to design a realistic, manufacturable water bottle by applying parametric modeling techniques and solid design principles.

The design focuses on **ergonomics, capacity accuracy, structural strength, and aesthetic simplicity**, similar to commercially available reusable water bottles.

Objective

- To design a 1-litre capacity water bottle using Fusion 360
- To understand parametric modeling and feature-based design
- To apply real-world dimensions and symmetry in product design
- To create a clean and manufacturable 3D model

Software & Tools Used

- **Autodesk Fusion 360**
- Sketch tools (Line, Circle, Arc)
- Extrude, Revolve, Fillet, and Shell features
- Parametric dimension constraints

Design Methodology

Base Sketch Creation

- A 2D sketch of the bottle profile was created on the front plane
- Dimensions were defined parametrically to ensure accurate proportions
- The profile included the base, body curvature, neck, and mouth

  Design Reference:
  Screenshot of the initial bottle profile sketch with dimensions.

Revolve Operation

- The 2D profile was revolved around a central axis

- This created a symmetrical 3D bottle body
- Revolve ensured uniform thickness and accurate cylindrical geometry

Feature Used: Revolve (360°)

Hollow Structure (Shell)

- The bottle was hollowed using the **Shell** feature
- Uniform wall thickness was applied to simulate real plastic bottles
- This step ensured material efficiency and realistic manufacturing design

Feature Used: Shell

Neck and Cap Area Design

- The bottle neck was designed to match standard bottle openings
- Smooth transitions were created between the body and neck
- Fillets were added to avoid sharp edges

Feature Used: Fillet

Final Finishing

- Rounded edges were applied for safety and aesthetics
- The base was slightly thickened for stability
- The final design was inspected for symmetry and proportion

Capacity Consideration

- The internal volume was designed to approximate **1 litre (1000 ml)**
- Dimensions were chosen based on standard bottle proportions
- The model reflects realistic size and usability

Learning Outcomes

- Gained hands-on experience with **Fusion 360 parametric modeling**
- Learned how to design **revolve-based products**
- Understood real-world product constraints such as wall thickness

- Improved spatial visualization and 3D design accuracy

Conclusion

The 1-litre water bottle project successfully demonstrates the application of **Fusion 360** for product design. By using sketch-based modeling, revolve operations, and shell features, a realistic and manufacturable bottle design was achieved.

This project strengthened understanding of **CAD fundamentals, parametric control, and industrial product modeling**, making it a valuable learning experience in 3D design.